

Determining the Saturation Point for UAV Operations in Airport Environments: A Probabilistic Approach

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Abstract—The rapid integration of Uncrewed Aerial Vehicles (UAVs) into airport operations has led to a new era of aviation challenges, particularly in ensuring safety and maintaining operational efficiency. The current protocol for obtaining approval for UAV operations within airport zones necessitates obtaining clearances from both Air Traffic Control (ATC) and relevant government bodies. However, this process is hindered by a lack of uniform standards, particularly concerning the maximum number of UAVs allowed to operate simultaneously, leading to inconsistencies and potential operational challenges. The paper addresses these challenges, focusing on the current approval processes for UAV operations within airport environments. Our study introduces a novel approach to calculate the saturation point for UAV operations in and around airports. This saturation point is defined as the maximum safe limit of UAVs operating, without causing disruptions or safety hazards. To accurately determine this threshold, we consider a comprehensive range of factors, including air traffic classes, the volume of crewed aircraft traffic, prevailing weather conditions, and the capabilities of the UAVs themselves. Our analysis divides the issue into two main risk categories: hazards arising from pre-approved UAV activities and risks related to crewed aircraft. We then propose a probabilistic approach that assesses the frequency of crewed aircraft traffic in conjunction with diverse UAV operational scenarios, such as building inspections, to calculate the saturation point (P_s). This model incorporates key parameters like geolocation, timing, and other operational metrics to predict potential mid-air collision (MAC) and near-mid-air collision (NMAC), as well as to ensure adequate clearance between aircraft. Furthermore, the paper outlines the essential operational parameters required from UAV operators and airport management to refine the accuracy of the saturation point calculation. To validate our approach, we use simulation supported by real flight data from Grand Forks International Airport, illustrating various scenarios and their impact on UAV operation safety. The outcome of this study is a probabilistic framework that provides an essential factor for air traffic control (ATC) and other relevant organizations to make decisions regarding the operation permission of UAV operation in and around airports. As a result, this research greatly contributes to the improvement of safe and efficient management of airspace in airport environments.

Index Terms—Unmanned Aircraft, Risk assessment, Airport operations, ATC management

I. INTRODUCTION

Uncrewed Aerial Vehicles (UAVs) have become crucial for enhancing airport operations, offering services to supplement

surveillance, security, construction oversight, airport inspection and facility maintenance. UAVs can provide real-time data to improve safety, minimize human error, and increase operational efficiency by conducting aerial inspections of runways, terminals, and perimeter fences [1]. Furthermore UAVs play a significant role in emergency response within an airport environment by capturing aerial imagery, thereby enabling fast, real-time decision-making. Moreover, these low altitude flying aircraft play a crucial role in wildlife management, reducing the risk of bird strikes and ensuring safer airport operations. In future, UAVs are expected to play a crucial role in optimizing airport operations and have the potential to become an integral tool in the aviation industry's to augment safety, security, and service quality.

The incorporation of UAVs into the shared airspace with crewed aircraft presents a multitude of challenges, that have the potential to impact the safety of flight. The main safety concerns include the risk of a MAC accident or an NMAC incident. The use of detect and avoid (DAA) systems which include, ground-based radar and similar technologies may help in reducing the risk of collision and improve operational efficiency. To ensure safety and security of the crewed aircraft operations, government bodies like the FAA enforce strict regulations for flying UAVs near airports. This includes the need to get permission from Air Traffic Control (ATC) before flying a UAV in near airports [2]. The primary safety concerns in an airport environment are the potential for a MAC accident or an NMAC incident, where crewed or uncrewed aircraft come very close to the operational volume of another aircraft. Various factors influence the hazards when UAVs operate alongside crewed aircraft; these include the operational volume, the size and speed of the aircraft, the maneuverable capabilities of the aircraft, etc. The development of a sophisticated air traffic management systems is necessary to effectively manage both crewed and uncrewed aircraft safety. For the safe integration of UAVs in an airport environment, systems are required that can effectively oversee the flight paths of both UAVs and crewed aircraft, thereby guaranteeing a safe operational environment. The number of UAVs operating in an airport plays a critical role and is influenced by multiple factors,

such as the volume of crewed air traffic, approved UAV operations, and the presence of critical infrastructure. This paper focuses on how these factors can be captured and analyzed for risk assessment, when ATC receives a UAV operational approval request from an operator. We believe this study is an initial attempt at evaluating the number of UAV operations in an airport environment. The study highlights the difficulties that come with incorporating, executing, and managing these activities in order to guarantee safety.

The current process for obtaining approval for UAV operations within airports often involves acquiring clearances from both the ATC and similar government agencies. This procedure, however, faces challenges due to the absence of uniform standards, and the risk associated with the number of UAVs allowed to operate simultaneously in the same area as crewed aircraft. This inconsistency in standards and lack of understanding of the risk associated with the number of UAVs allowed to operate simultaneously in the same area as crewed aircraft leads to potential operational challenges, as there is a lack of a clear, standardized approach to managing UAV activities within the sensitive and complex environments of airport operations. The paper tries to address this problem by analyzing the risk associated with UAVs operating in and around airports. The maximum number of UAVs allowed to operate (*Saturation value*) as defined in this paper, represents the limit on the number of UAVs that can safely operate without increasing risk. Determining this threshold is crucial for ensuring the safety and efficiency of both UAV and crewed aircraft operations, as it helps prevent potential mid-air and near-mid-air collisions. The proposed approach divides risk analysis into two main categories: (1) risks arising from pre-approved UAV activities and (2) risks related to crewed aircraft. These highlight the importance of understanding these distinct risk profiles to manage UAV operations effectively around airports. This differentiation is crucial for developing targeted strategies to mitigate specific risks associated with UAV integration into airport environments. Furthermore, the study uses a probabilistic risk assessment that allows the calculation of the saturation value by predicting potential mid-air and near-mid-air collisions, thereby ensuring adequate clearance between aircraft. This approach underlines the significance of accurately understanding and managing the risks associated with UAV operations to enhance safety and operational efficiency in the airport environment.

The remaining sections are organized as follows. Section II covers the background information. In Section III, we describe the methodology and the various metrics applied in our analysis. Section IV presents the simulation scenarios and their outcomes. Finally, Section V summarizes the findings of this study and the future scope.

II. BACKGROUND

The number of UAVs that can be operated in an airport environment require a standard analysis process that must be addressed to ensure the safety, and efficiency of both crewed and un-crewed aircraft operations. Without specific standards

to identify UAV saturation levels within a pre-determined area, such as the airport environment present the increased likelihood of operational conflicts and safety incidents. This emphasize the urgent need for regulatory/policy frameworks that accommodate the unique characteristics and operational needs of UAVs within the complex ecosystem of airport operations.

The FAA published an article discussing the management of crewed aircraft operations and the assessment of airport capacity [3]. The article discusses several factors that are considered, including as runway utilization, aircraft maintenance, arrival and departure procedure, taxiway designs etc. These combined parameters determine the calculation of the hourly capabilities for both Visual Flight Rules (VFR) and Instrument Flight Rules (IFR). The Annual Service Volume (ASV) is an important measure used to estimate the yearly capacity of an airport, taking into account factors such as runway utilization, types of aircraft, and variations in weather conditions. In addition, aviation delay is computed by taking the average service time (ASV) and comparing it to the expected annual demand.

Previous studies analyzed the risks associated with UAVs in proximity to crewed aircraft, considering the ground risk assessment and air collision risk assessment. Studies propose dynamic models based on deep learning for assessing ground risk by considering factors such as population density, ground impact, shelter, etc [4]. Studies on UAV air collision risk with crewed aircraft include research on collision avoidance systems. These systems optimize UAV trajectories to avoid obstacles and other aircraft, focusing on adherence to performance and operational constraints. Local trajectory planning algorithms proposed in [5] demonstrate successful avoidance maneuvers. Local trajectory planning algorithms proposed in [5] demonstrate successful avoidance maneuvers, enabling UAVs to return to global trajectories post-maneuver .

The study conducted by [6] investigated the feasibility of employing UAV for logistic delivery, emphasizing the critical role of supporting technologies and risk assessments. The study employs an Expected Level of Safety (ELS) analysis to address safety concerns for UAVs. Another study, [7] proposes a risk-based methodology to determine the capacity of UAV only corridors, considering third-party risks on the ground. The research underscores the significant impact of UAV failure rates on corridor capacity and suggests management strategies for fixed-corridor applications such as package delivery. These studies, however, have not considered the UAV risk associated with crewed aircraft.

The work presented by [8] investigated post-accident analysis, benchmarking, and improvements related to uncrewed vs. crewed aircraft systems accidents. The work also provided insights into understanding collision risks and enhancing safety measures[9] described a safety analysis methodology vetted by the FAA, Eurocontrol, and the International Civil Aviation Organization (ICAO). The paper focus on assessing UAV collision avoidance system performance. A 3D collision risk assessment model is presented in [10] for scenario when

a non-cooperative UAS and UAS operating within restricted airspace. This study assessed both the collision risk and the safety risk associated with UAS.

The risk analysis of UAV operations around airports was carried out in [11]. The study proposes a tool to evaluate the probability of collisions between UAS and crewed aircraft. The framework specifically focuses on flights below 1000 ft altitude in and around airports. The study intends to assess the risks of MAC, NMAC, and Well Clear violations in busy airport areas by using historical Automatic Dependent Surveillance–Broadcast (ADS-B) data to simulate aircraft paths and adopting a probabilistic method to predict UAS trajectories. The analysis highlights the specific times when Grand Forks International Airport experiences the highest level of activity. This demonstrates the possibility of disruptions and economic losses. It emphasizes the significance of conducting thorough risk assessments to ensure the safety and efficiency of both uncrewed and crewed aircraft operations at and around airports.

III. PROPOSED APPROACH

When ATC receives a request to operate a UAV, a risk assessment should be performed to decide whether the operation can be approved. It is crucial for agencies to determine the optimal number of UAVs that an airport can safely manage. This evaluation requires an analysis of both current crewed aircraft operations and any previously authorized UAV activities within the airport’s jurisdiction. It is vital to analyze each instance separately to accurately assess the risk associated with the operation. The approval of UAV operations depends on the estimated risk probabilities, which must align with predefined risk criteria. This value is airport-dependent, as it relies on the aircraft traffic and related criteria that an airport can handle. Such a systematic approach is essential for ensuring the smooth and safe operation of UAVs alongside crewed aircraft, highlighting the importance of rigorous risk management in air traffic control protocols.

For analysis, it is crucial to have a clear understanding of the parameters that govern the operation of UAVs, such as their speed, altitude, and assigned waypoints, in order to appropriately assess the risks involved. An essential aspect of this assessment is evaluating variables such as the likelihood of a MAC, NMAC, or well-clear violation, which have significant impacts on safety levels. Unlike the operators of crewed aircraft, UAV operators do not necessarily maintain continuous communication with ATC, which presents unique challenges in managing risks. Furthermore, the potential for UAV operations to pose hazards to infrastructure, especially critical infrastructure within an airport environment, increases the overall level of risk. The importance of a thorough risk analysis framework that includes both crewed aircraft and UAV operations is emphasized to ensure the safety and reliability of airport operations.

The analysis incorporates various parameters, such as the risk volume associated with crewed aircraft and UAVs, the trajectories of both crewed aircraft and UAVs, and the spatial

dimensions of the airport area, among other factors. The following section provides a detailed description of these factors, focusing on the several complex factors that affect the safety and efficiency of incorporating UAVs into airspace that are also used by crewed aircraft. *Assumptions:* The approach proposed evaluates the interaction between UAV operations and crewed aircraft traffic using probabilistic techniques. The approach assumes that it focuses on scenarios such as building inspections and linear inspections to estimate collision risks. The paper also assumes the maximum altitude of operation is limited to 100 ft to ensure safe and secure operations. By UAVs, the authors assume the use of small-UAVs throughout the paper. The paper also assumes that all UAVs have a built-in remote ID(RID) where other operators can detect an intruder’s presence.

A. Risk Volumes Associated with Crewed Aircraft

The paper uses three risk volumes associated with crewed aircraft: Mid-Air Collisions (MAC), Near Mid-Air Collisions (NMAC), and Well Clear (WC) [12] (refer to Figure 1). The dimensions for NMAC and WC volumes are derived from previous studies [12], [11], while MAC volumes are determined based on approximate aircraft dimensions in a 3D space. The spatial risk volume for WC of crewed aircraft is defined by a radius of 2000 ft and a height of 500 ft (i.e., 250 ft above and 250 ft below), as established in prior research [12]. Specific details for each cylindrical volume are provided in Table I.

B. Risk Volumes Associated with un-crewed Aircraft

Similar to crewed aircraft, it is crucial to define a risk volume when multiple UAVs are operating in the same area in order to avoid any collisions. Multiple studies in the literature suggest various risk thresholds for the operation of many small UAVs, emphasizing the range of safety precautions needed for such situations. We employ the RTCA DO-396 small-Near Mid-Air Collision (sNMAC) volume criteria in this work, which requires each UAV to keep a distance of 50 feet horizontally and 15 feet vertically from other UAVs [13]. This rule guarantees a uniform safety buffer, which decreases the likelihood of UAVs coming too close to one another or colliding when they are in the same airspace.

TABLE I: Risk volumes and their dimensions considered in the present study. NMAC and WC height reflects volume of space not distance from aircraft – UAS will remain 250 ft vertically clear of a crewed aircraft

S.No	Risk volume	Radius (ft)	Height (ft)
1	Mid-air collision (MAC)	40	20
2	Near mid-air collision (NMAC)	500	200
3	Well-clearance (WC)	2000	500

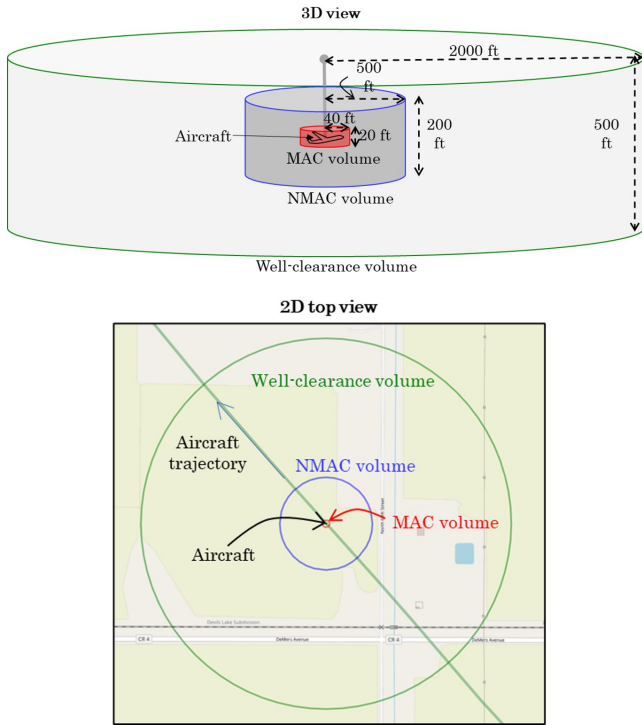


Fig. 1: Risk Volume - Crewed

C. Crewed Aircraft Trajectories

The acquisition of aircraft traffic information entails the extraction of data from the ADS-B system. The dataset utilized for analysis spans the temporal interval between April and September 2021, sourced from the ADS-B receiver deployed at the Grand Forks International Airport, as provided by Flightradar24 [14]. Comprehensive pre-processing of the ADS-B data is imperative to isolate crewed air traffic within the designated volume of interest proximate to the airport. This pre-processing endeavor encompasses the elimination of stationary traffic instances, such as those stationed on the ground without motion, and the interpolation of absent data points.

The ADS-B data may exhibit non-uniform sampling, necessitating the interpolation of intermediate points to complete the temporal resolution and depict the dynamics of aircraft maneuvers, including turns, ascensions, and descents onto runways. The selection of the sampling interval is contingent upon the specific analytical objectives and the requisite precision for probability computations. In the context of the presented research, the ADS-B data has undergone re-sampling at a uniform 1-second interval.

The resultant dataset derived from ADS-B records encompasses key flight-specific parameters, including i) temporal information, ii) geospatial coordinates (latitude and longitude), and iii) altitude. Notably, the determination of altitude above ground level (AGL) is executed by utilizing the ADS-B altitude data in conjunction with digital elevation maps procured from the United States Geological Survey (USGS) EarthExplorer database [15]. Figure 2 shows the ADS-B

monthly statics.

Our observations reveal that the busiest air traffic occurs mostly on Tuesdays in June. Figure 3 illustrates that the highest number of flights takes place between 10:00 to 11:00 AM CT on Tuesdays. To further analyze, we focus on flight trajectories between 10:00:00 to 10:22:00 (CT), as shown in Figure 4

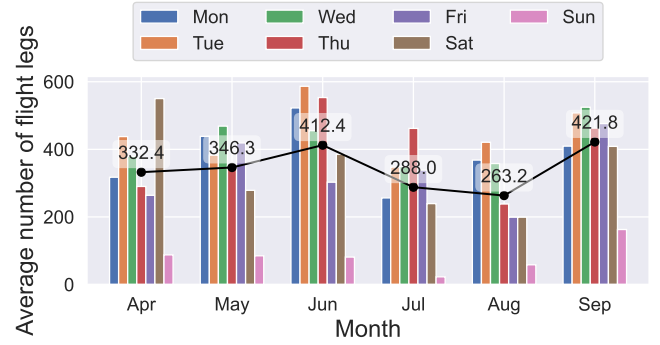


Fig. 2: Statistical histograms showing the average flight segments within the Grand Forks International Airport (GFK). The data is sourced from flightradar24.com [14].

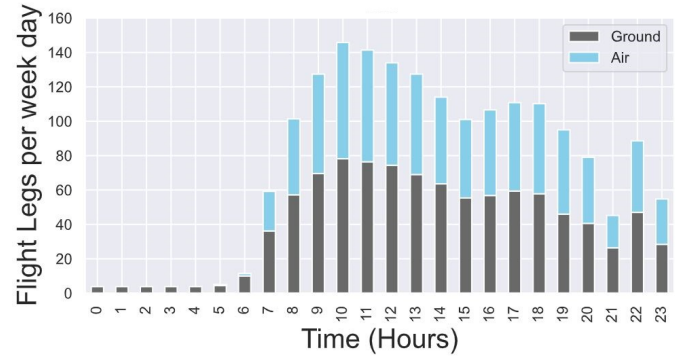


Fig. 3: Average flight segment trends plotted every hour on Tuesdays at the Grand Forks International Airport (GFK). The data is sourced from flightradar24.com [14].

D. UAV trajectories

Each UAV trajectory is defined by an origin, end point, and elevation. These parameters are randomly sampled from a uniform distribution, adhering to two crucial conditions: i) Both the starting and ending locations must fall within the user-defined restricted airspace, and ii) the distance between the origin and end locations always equals the user-specified segment length.

In this study, we employed a restricted airspace modeled as a cylinder with a radius of 2 miles and an elevation of 1000ft. Each UAV segment has a fixed length of 1000ft, and the elevation for each UAV is consistently set at 20 feet. It is essential to highlight that while we utilized a

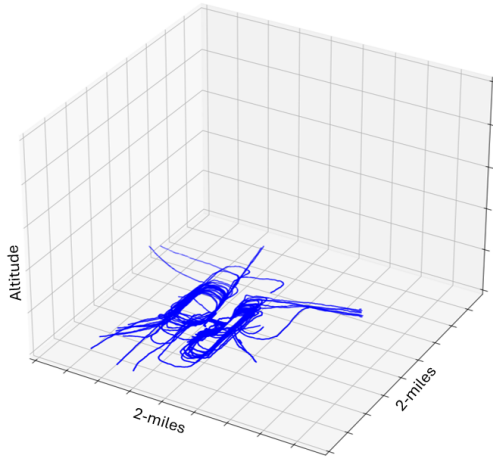


Fig. 4: Figure shows a 3D view of crewed aircraft trajectory for a peak hour operation from GFK airport

specific cylindrical volume and a constant segment length to demonstrate the study's concept, this approach can seamlessly adapt to arbitrary shapes and lengths, as well as varying elevations for the UAV trajectory.

E. Risk Analysis Quantification

In the preceding sections, we delineated the trajectories of both aircraft and UAVs. The subsequent phase involves identifying and quantifying the potential risks associated with UAVs entering the risk volumes of crewed aircraft. The probability of UAV-related risks concerning aircraft can be computed in either the temporal or spatial domain. In our analysis, we have opted for the temporal and spatial domains, depending on the scenario.

For the given set of flight trajectories T , representing aircraft landings and takeoffs at the airport. For each trajectory $t_i \in T$, a MAC volume is identified based on the dimensions of the aircraft. Let the dimensions of the aircraft be represented by a volume V_{MAC} .

Similarly, the NMAC volume (V_{NMAC}) for interaction with a crewed aircraft is defined as a cylinder with a horizontal radius of 500 ft and a vertical range of 100 ft for each trajectory t_i .

The Well Clear Violation volume (V_{WC}) for each trajectory for interaction with a crewed aircraft is defined by a horizontal distance of 2000 ft and a vertical distance of 250 ft.

For a proposed UAV operation at location $L_j \in \{L\}$, the safety requirement is such that L_j must not intersect with $V_{MAC}(t_i)$, $V_{NMAC}(t_i)$, or $V_{WC}(t_i)$ for any $t_i \in T$, ensuring the UAV operation does not pose a risk of collision or well-clear standard violation. For each trajectory, the corresponding volume is calculated.

In this approach, the duration of travel for each UAV is first calculated, and the instances when it might encroach upon the risk volume are recorded. The risk probability is subsequently determined by dividing the potential risk time by the total duration of the UAV's trajectory. Time is uniformly sampled

at 1-second intervals for both UAVs and aircraft trajectories. To be more explicit, the probability associated with each risk volume is calculated as follows:

$$P(MAC) = \frac{t_{MAC}}{t_{TOTAL}} \quad (1)$$

$$P(NMAC) = \frac{t_{NMAC}}{t_{TOTAL}} \quad (2)$$

$$P(WC) = \frac{t_{WC}}{t_{TOTAL}} \quad (3)$$

Where t_{TOTAL} represents the total time duration of the UAV. Additionally, t_{WC} denotes the time during which the UAV remains within the Well-Clear (WC) volume of the aircraft, t_{NMAC} signifies the duration the UAV spends within the NMAC volume, and t_{MAC} characterizes the time the UAV occupies within the MAC volume of the aircraft. The visual representation of these temporal intervals, along with the corresponding risk volumes of the aircraft, is depicted in Figure 5.

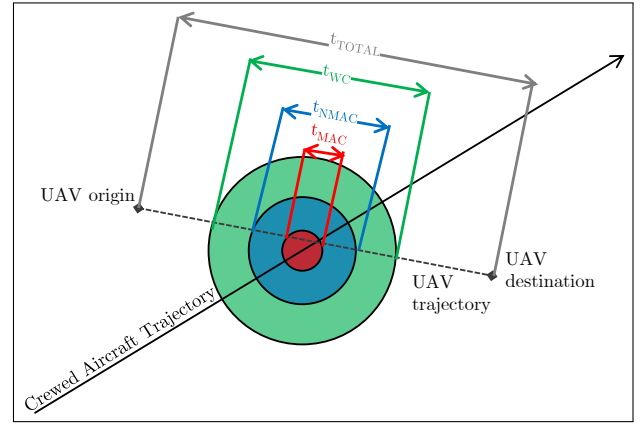


Fig. 5: Schematic illustrates UAV temporal intervals: t_{TOTAL} for total duration, t_{WC} within Well-Clear (WC) volume, t_{NMAC} in Near Mid-Air Collision (NMAC) volume, and t_{MAC} within Mid-Air Collision (MAC) volume

F. Quantification Metric for UAV saturation Limit

In preceding investigations, researchers investigated the incremental risk associated with the integration of UAVs into the designated restricted airspace. [16] having introduced a mathematical approach to gauge airspace capacity. Specifically, concentrating on the influence of uncrewed aircraft systems in low-altitude operations. Based on the work, [17] proposed a methodology for evaluating the operational capacity of a UAV logistics route network within the realm of urban air mobility (UAM). They developed a bi-objective optimization model, striving to maximize executed UAV logistics delivery plans while minimizing the total operational impedance value. Parallel studies [18], [19] primarily focused on estimating capacity within the urban air mobility context and did not specifically address UAV capacity within the airport.

The paper propose a metric primarily tailored to aircraft operations within the airport scenario. This metric, denoted as (S), can be mathematically defined as

$$S = \frac{1}{N} \sum_{i=1}^N [\alpha P_i(MAC) + \beta P_i(NMAC) + \gamma P_i(WC)] \quad (4)$$

where N represents the total number of approved UAVs along with the new request the ATC receive for operating within the airport, and α , β , and γ are weigh factors assigned to the probabilities $P_i(MAC)$, $P_i(NMAC)$, and $P_i(WC)$, respectively. These coefficients are contingent upon the risk tolerance levels specific to each airport.

It is crucial to highlight a departure from the approach taken by earlier studies, where UAVs are maneuvered to avert collisions, and the final capacity metric relies on user-defined minimum thresholds. In our study, we calculate the anticipated risk before any maneuver occurs. This research aids in identifying the projected risk associated with incremental UAVs. While successful detection, conflict resolution, and safe UAV maneuvers have the potential to eliminate risks, our focus lies on preemptive risk assessment.

We conducted parametric studies on the weighting coefficients α , β , and γ , as well as the number of UAVs (N). Subsequently, we explained the impact of each of these parameters, detailing the approaches for selecting an appropriate threshold value for S .

G. Multiple UAV operation

The scenario when ATC receives a new UAV operation request requires further analysis to estimate the risk associated with other UAVs operating in the same environment. This includes scenarios when different UAVs operate in the same space and the waypoints of different UAVs overlap. An analysis was made to estimate the possibility of risk when UAVs operate in the same space. For this paper, we assume a linear inspection of UAVs flying in a straight line. Since there are multiple UAVs in the same trajectory, it is also important to follow Right of way (RoW) rules while operating in close proximity. RoW rules for UAVs are essential protocols designed to prevent collisions and ensure safe airspace sharing. These rules dictate how UAVs must yield to crewed aircraft and other UAVs under specific circumstances, prioritizing safety and efficient traffic flow. Understanding and adhering to RoW rules are crucial for operators to navigate complex airspaces and maintain operational integrity.

As per the FAA regulations for crewed aviation, the aircraft on the right has the right of way when two or more aircraft are converging at approximately the same altitude. As per FAA regulation, all uncrewed aircraft must yield the right of way to crewed aircraft. For operations beyond the scope of Part 107, such as those under special waivers or within controlled airspace, the FAA may impose additional requirements or operational conditions to ensure safety and compliance with the National Airspace System (NAS) rules.

When two UAVs encounter each other, the emphasis would be on maintaining safety and preventing a collision, likely applying a similar logic to the general right-of-way rules found in crewed aviation—such as the UAV to the right having the right of way or both operators taking action to avoid a potential conflict.

For analysis, we consider a worst-case scenario where the trajectory of UAVs is between buildings. In such a situation, an UAV may not be able to follow the usual RoW rules, such as overtaking via right. In this paper, we assume that when multiple UAVs operate and cross over each other, we propose a vertical overtaking rather than a horizontal overtaking. For the vertical and horizontal separation, we consider the protection volume for sUAS intruders, which is 50ft horizontal and 15ft vertical "hockey puck" volume, which is called sUAS Near Mid-Air Collision (sNMAC) and is analogous to the NMAC volume for crewed aircraft.

H. Risk associated with GPS uncertainty

Analysing GPS uncertainty is important to evaluate the risk associated while operating UAVs in airport particularly near buildings where Dilution of Precision(DoP) value is high. As per the guidelines outlined by the ICAO [20], Position, Navigation, and Timing (PNT) solutions are defined by their accuracy, integrity, continuity and availability all of which align with the principles of the Performance Based Navigation (PBN) framework (figure 6). PNT data play a central role in guidance and control systems. The Global Navigation Satellite System (GNSS) is a primary PNT source, known for its extensive coverage, precision, and cost-effectiveness. Thus, vigilant monitoring of GNSS performance becomes pivotal for safe and reliable UAV operations [21].

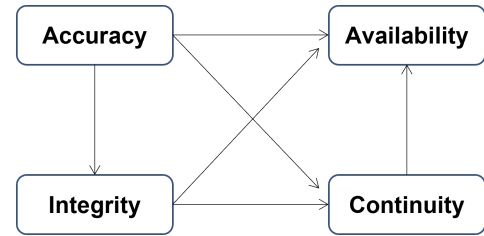


Fig. 6: Performance Based Navigation, based on [21]

Section IV presents an overview of the impact of GPS and DoP on UAV operations.

IV. RESULTS

This section presents the findings based on an analysis of ADS-B data, which tracks flight traffic within a 2-mile radius of Grand Forks International Airport (GFK). Simulations of UAV trajectories were conducted to illustrate how UAVs navigate in and around the airport. Separate simulations were carried out to evaluate the risks associated with operating an increasing number of UAVs alongside crewed aircraft, as well as scenarios where only UAVs are present.

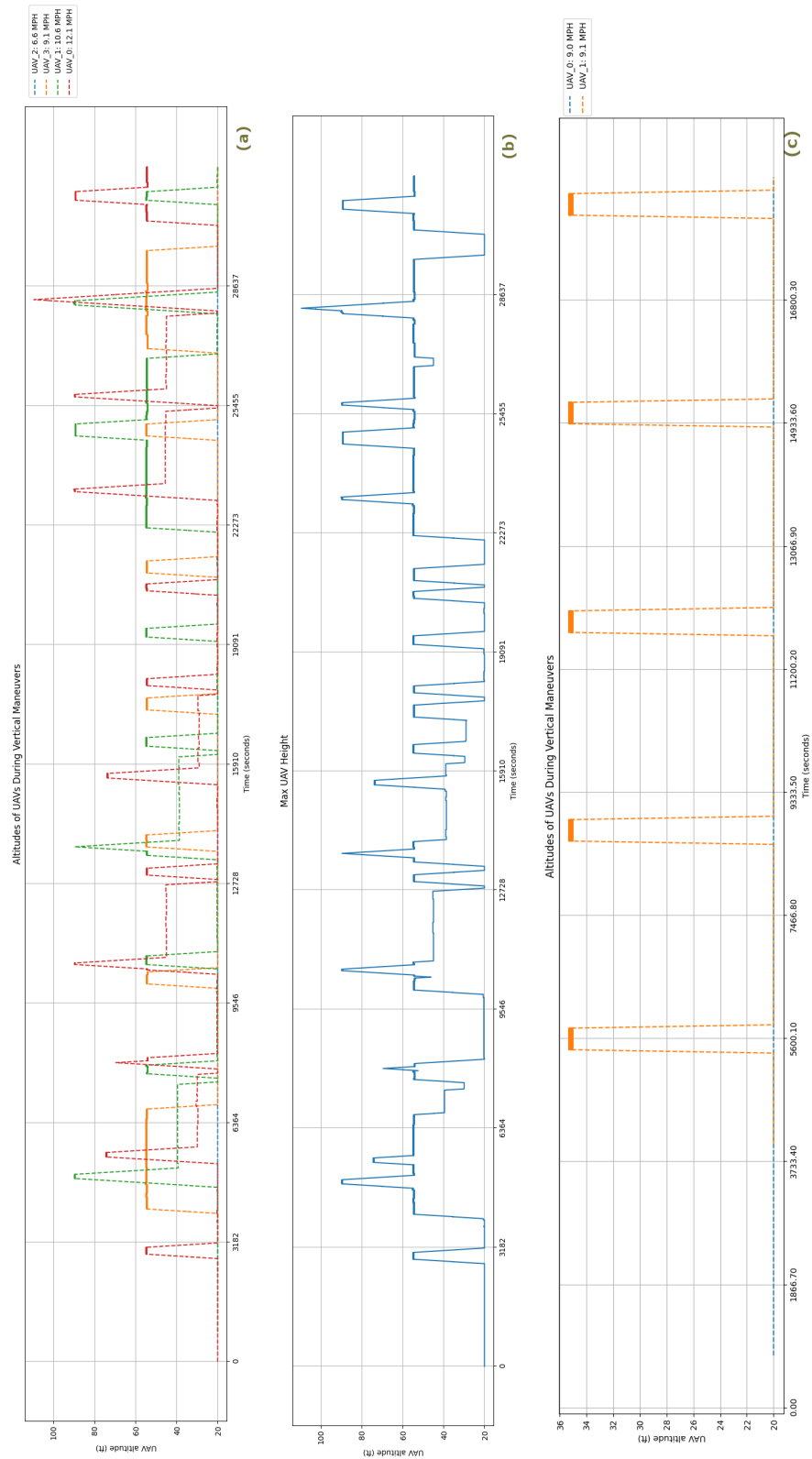


Fig. 7: Captions for Images. (a) Scenario of four UAVs inspecting a building in a loop. (b) The overall max-altitude of operations with four UAVs. (c) The scenario of two UAVs inspecting in opposite direction around the building.

The number of UAVs operating in the same spatial area should be carefully controlled to prevent potential hazards. simulation scenarios were designed to have multiple UAVs sharing the same waypoint. Although multiple simulation scenarios were executed, this paper highlights a specific use-case that shows a worst-case scenario where UAVs cannot maneuver horizontally. The aim is to highlight the importance of additional safety precautions when multiple UAVs, managed by different operators, share airspace. The scenario presented in the paper assumes that the multiple UAVs are navigating around the same infrastructure, with varying speeds and directions, to illustrate the complexities and risks involved.

In our study, all UAVs are programmed to fly at a uniform altitude of 20 feet. As they approach each other, maintaining the sNMAC volume becomes important. The scenario highlighted in the paper focuses on UAVs navigating through urban environments, specifically flying between buildings, where vertical maneuvers become the only feasible option for uninterrupted operation. This raises a challenging question for operators: determining which UAV should execute the vertical maneuver.

To address this issue, we recommend the development of an intelligent, interconnected UAV system. Such a system would facilitate the centralized decision-making necessary for coordinating these maneuvers efficiently. The illustrations in our paper depict a scenario where multiple UAVs navigate toward the same waypoint, each at varying speeds and directions. A continuous simulation loops was done to determine the saturation point (maximum altitude) for multiple UAVs operating simultaneously within a confined space.

Figure 7.a shows four UAVs operating in the same space and their interaction. Figure 7.b shows the maximum altitude the UAVs reach over time. Given the operational altitude of 20 feet and a maximum allowable altitude of 100 feet, we conclude that the maximum feasible number of UAV operations in this scenario is four. Additionally, we present a scenario involving two UAVs operating in the same space, further emphasizing the complexity and potential risks of such operations, as shown in figure 7.c.

The scenario of a GPS inaccuracy is explained below. Position accuracy can be attributed to the extent to which the computed or observed location and/or speed of a UAV at a specific instance aligns with its actual location or speed. Low GPS accuracy can be attributed to several factors. Geometric/Position Dilution of Precision (GDOP/PDOP) plays a role by indicating how satellite positions relative to each other affect the accuracy [22]. Atmosphere refraction can bend signals, leading to signal delays. Multi path effects occur when signals reflect off surfaces before reaching the receiver, causing signal interference [23]. Satellite time and location accuracy influence precision, as inaccuracies impact signal calculations [24], although this has been largely mitigated. These combined factors contribute to variations in GPS accuracy, emphasizing the need for advanced techniques to enhance overall positioning precision [25].

The combined error in receiver placement consists of the

User Equivalent Range Error (UERE) and the Dilution of Precision (DOP). The formulation for the standard deviation of the receiver's positioning error, σ_p is as follows [26]

$$\sigma_p = DOP \times UERE \quad (5)$$

The user equivalent range error (UERE) denotes the variations observed during the process of measuring the distance between a receiver and a satellite. These UERE disparities are quantified as errors encompassing both positive and negative values, indicating a lack of bias or a zero-mean characteristic [27].

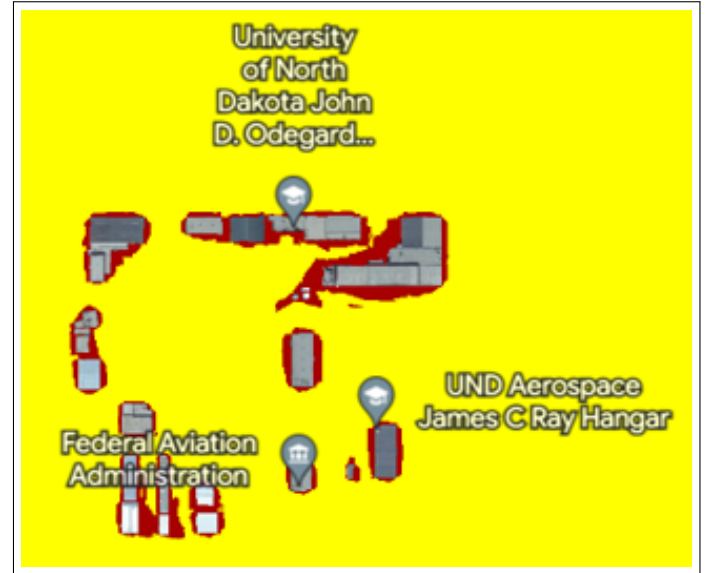


Fig. 8: GPS uncertainty maps in proximity to GFK airport, showcasing PDOP values. Red indicates $PDOP > 5$, while yellow depicts $2 < PDOP < 5$. Image process from the data provided by Spirent Communications plc.

Figure 8 shows the GPS-DoP coverage at a low altitude at a random time from GFK airport. The red line around the building shows the GPS coverage and indicates a DoP value near to 20. This basically indicates the possibility of hazards when UAVs are operated in close proximity to building or between buildings. In a similar situation, the sNMAC volume may not be safer for a close-proximity operation and may need further evaluation.

Figure 8 illustrates the GPS Dilution of Precision (DoP) coverage at low altitude in a worse case near GFK airport. The red line encircling the building represents the GPS coverage area, indicating a DoP value close to 20. This highlights the potential risks of operating UAVs in tight spaces, such as between buildings or obstacles, where the sNMAC volume might not provide a safe margin for close-proximity operations. This scenario suggests a need for further evaluation to ensure safety in such complex environments.

The study also evaluates the volume of crewed aircraft traffic at GFK Airport by leveraging actual traffic data for comparison. To accomplish this, peak traffic times at GFK

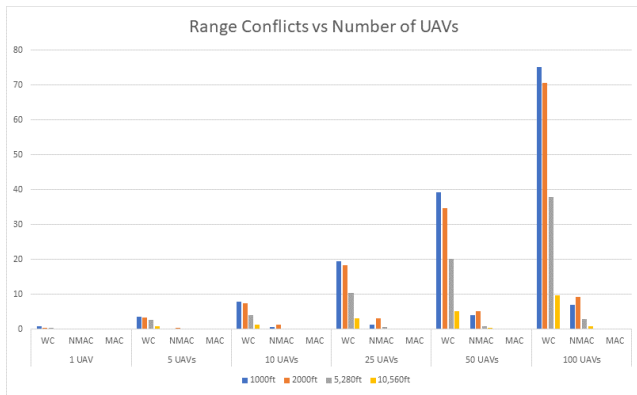


Fig. 9: Range Vs Risk Volume

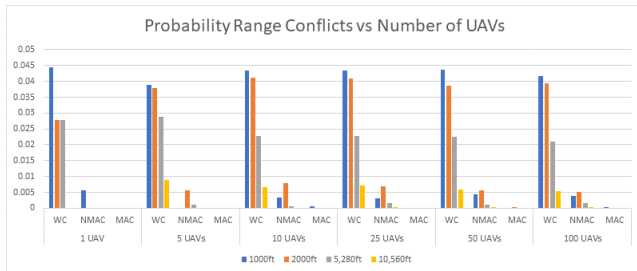


Fig. 10: Probability Vs UAV operations

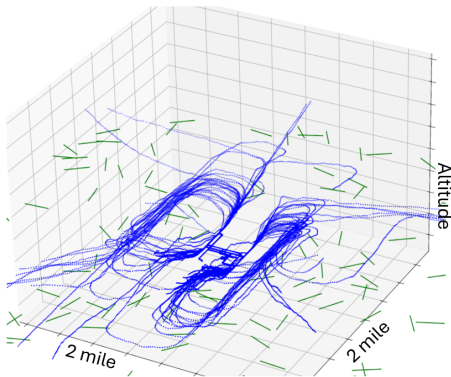


Fig. 11: Two mile radius around the airport with multiple UAVs operations. Blue lines indicate the crewed aircraft traffic, Green lines indicate the simulated UAV trajectories

Airport were analyzed alongside simulated UAV traffic patterns. A variety of UAV flight paths were integrated within a 1,000-foot to 2-mile radius of the airport. These UAV paths varied in quantity, from 1 to 100, and explored a range of distance values to assess their impact on the airspace. For each trajectory, the interception of MAC, NMAC, and WC volumes was analyzed to calculate the number of UAVs that fall under each scenario. The average of multiple iterations was calculated to capture the simulation results.

Figure 9 shows the number of MAC, NMAC, and WC instances for each trajectory. The UAVs were operated at a 20ft altitude and at a linear horizontal distance of 1000ft. The

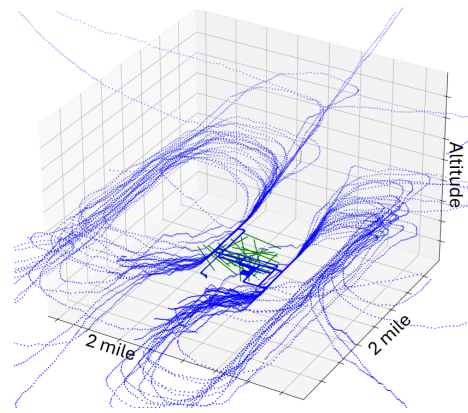


Fig. 12: Simulation of few UAVs operating in with in a 1000ft radius of airport. Blue lines indicate the crewed aircraft traffic, Green lines indicate the simulated UAV trajectories

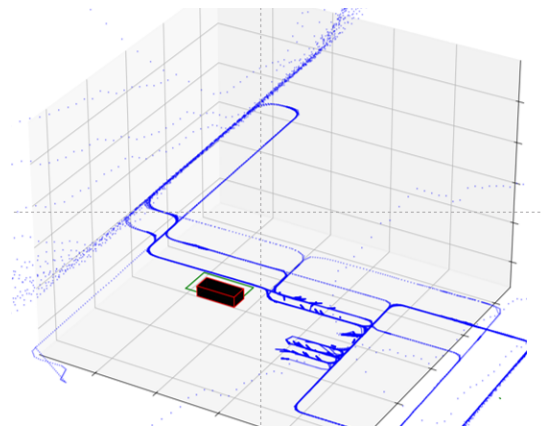


Fig. 13: A close view of UAV waypoints around a building at GFK airport, along with the crewed aircraft traffic (blue lines)

figure shows that with single UAV operations, the incidence of overlapping with the collision volume is minimal. But with the increasing number of UAVs, the WC and NMAC range gradually increases. A MAC instance is captured when 100 UAVs operate at close range. Though this approach gives an overall count of the number of UAV operations in an airport and its impact on the crewed aircraft's collision volume, it doesn't give values to identify how each new UAV operation has an impact on it. Figure 10 shows the probability of MAC, NMAC, and WC when UAVs operate in and near airports. This is calculated with the total distance(or time) the UAV spends in these collision volumes. A similar calculation can be used when an operator submits a new request to ATC.

Figures 11 and 12 show the crewed aircraft traffic and the UAV trajectories. Figure 11 shows the distribution of multiple UAV operations within a two-mile radius of the airport. The blue line indicates the crewed aircraft trajectory, and the green line indicates the UAV trajectory. Figure 12 shows a few UAVs operating in a proximity range near the center of the airport. A real-world scenario of a UAV inspecting a building is shown in figure 13, where crewed aircraft traffic is vertically

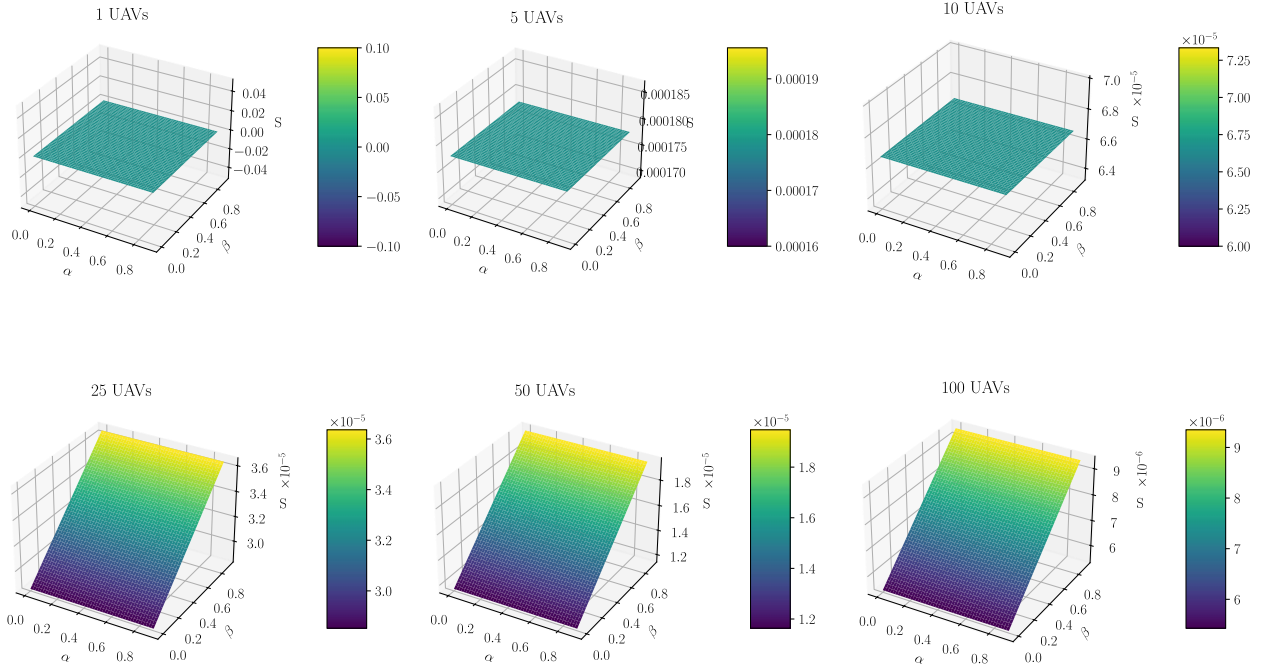


Fig. 14: Saturation metric dependence on the α , β and the number of UAVs, the γ value is fixed at 0.1. This specific case is corresponding to 10560 ft. α , β values are plotted on the X and Y axis, and the metric value (S) is presented on the Z axis.

above the UAV operation. For each waypoint of the UAV operation, the values of $P(MAC)$, $P(NMAC)$ and $P(WC)$ will be calculated, and the corresponding value of S will be calculated. This value of S depends on the values of α , β and γ . The size of the collision volumes is inversely proportional to their range. This means that if $P(MAC) > 0$ should be given more care in approving the operation than $P(NMAC)$. This means $\alpha > \beta > \gamma$. A parametric analysis is given to show how these constant values change with different operational scenarios.

A. Parameter Sensitivity Analysis

The influence of α and β on the saturation metric is delineated in Figure 14. This investigation maintains γ at a constant value of 0.1 while allowing α and β to range from 0 to 0.9, ensuring their collective sum with γ remains invariant at 1. It becomes evident that the saturation metric exhibits a notably higher sensitivity to variations in β compared to α . This phenomenon arises due to the markedly elevated probability of NMAC relative to MAC. Furthermore, the sensitivity of γ surpasses that of α and β , reflecting a higher incidence of WC scenarios compared to both NMAC and MAC events. This disparity underscores the inadequacy of a first-order equation for the saturation metric solely dependent on $P(MAC)$, $P(NMAC)$, and $P(WC)$ to yield unique metric values. Notably, even a probability as low as 10^{-6} for $P(MAC)$

retains significance, particularly in airport-related operations. Understanding these intricacies is paramount in formulating a comprehensive formula that accounts for MAC, NMAC, and WC parameters. The limitations explained in this study will definitely be included in future work that aims to create a mathematical models for saturation metrics.

V. CONCLUSION

In conclusion, the determination of a safe operational threshold for UAVs within airport zones is at-most important to ensuring the safety and efficiency of aviation operations. This study addresses the complexities of this challenge by analyzing the risk assessments for both pre-approved UAV activities and interactions with crewed aircraft. When multiple UAVs interact with each other, a safety risk volume proposed in RTCA DO-396 is considered for analysis. Similarly, when evaluating the risk of UAV operations associated with crewed aircraft, the probability of mid-air collisions (MAC), near-mid-air collisions (NMAC), and well-clear risk volume were analyzed. The number of UAV operations was captured using these three risk volumes to execute the count of MAC, NMAC, and WC occurrences with different volumes of operations. We propose a combination of three weighted constants that can be used for calculating the saturation value. Even though the paper separately evaluates the risk posed in two different scenarios, there can be other situations where a detailed

analysis is needed. This paper will serve as an initial base for the saturation point discussion. Based on the results, we recommend that implementing an advanced connected UAV system could enhance operational safety when multiple UAVs are in use within the same airspace. A central decision making system can help in performing safe maneuverable operation than individual operators control UAVs. Additionally, we suggest that Air Traffic Control (ATC) be notified well in advance of any UAV operations approved to fly in or near airports. This advance notification would enable ATC to conduct real-time risk assessments based on factors such as flight traffic, GPS availability, the presence of other UAVs in the area, and weather conditions. In the future, the authors aim to conduct a detailed analysis of various scenarios and develop a framework that can identify the risks associated with different operational contexts. This will not only apply to UAV operations in airports but also to Advanced Air Mobility (AAM) and Urban Air Mobility (UAM) vehicles operating both in airports and restricted airspace

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